

Chapter 5

Storage Devices

Storage devices are used to store data permanently. Data is stored on them for future use. We have different storage media:

- ✓ Floppy disk
- ✓ Hard Disk
- ✓ Compact Disc (CD)
- ✓ Flash Disk

Each of these devices have a device that read data from them or write data onto them. This device is called drive

Floppy Drive

Floppy disk is made from a thin piece of plastic coated with magnetic material on both sides. The magnetic material is iron oxide most of the time. Data is written to disk by magnetizing and demagnetizing the magnetic material.

Floppy is divided into logical areas for storing data on them. These are called sectors and tracks. Tracks are narrow concentric circles on a disk. Sectors are pie shaped slice of individual tracks. Sectors form the unit for data storage. Data is written or read one sector at a time. A single sector can store 512 bytes. The division allows floppy to be random access device. Data can be read from any sector or track easily.

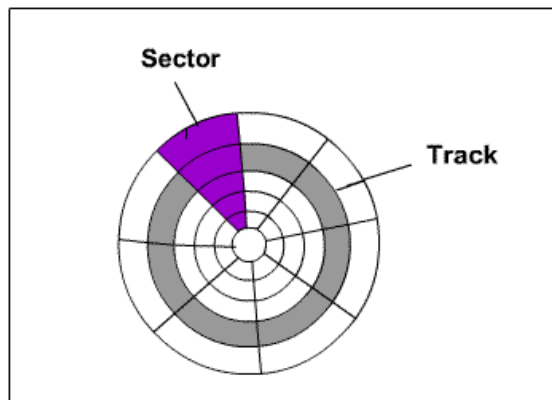


Fig. 1 sector and track of floppy disk

Note: the division is not visible for us. It is a logical division which is understood by computer only.

Data is stored on both sides of the disk .i.e. double sided. Early disks use only one side of the disk i.e. single sided.

Floppy Drive Components

Floppy drive contains several basic components. Some of the components are:

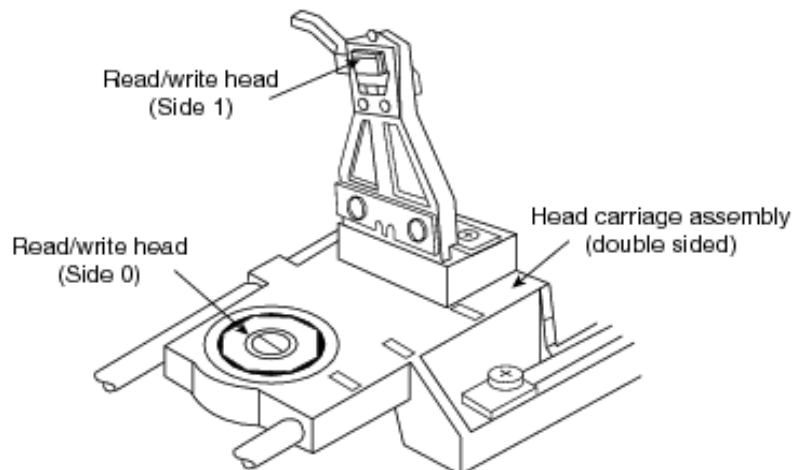
1) *Read Write Head*

Floppy drive normally has two read-write heads one for each side of the disk. Both heads are used for reading and writing on their respective disk sides.

Read-write heads convert binary data electromagnetic pulse when writing to disk. They do the reverse during reading i.e. convert electromagnetic pulse into binary data.

The head move in and out over the surface of the disk to position themselves over correct track and sector. It is moved by a motor called actuator.

The read-write head is made up of soft ferrous (iron) compounds with electromagnetic coils. To read or write data, the head must be in direct contact with magnetic medium. Very small particles of loose oxide, dust, dirt, smoke, fingerprints, or hair can cause problems with reading or writing the disk. Because of this, we have to handle the disk carefully.



2) Head Actuator

The head actuator is the device that physically position the read-write head over the correct track on the surface of the disk. Floppy disks contain 80 tracks per side. The actuator is driven by a motor called stepper motor. The motor turns a precise specified distance and stops, it does not spin around continuously. The stop positions are the tracks of disk.

3) The Spindle Motor

The spindle motor spins(rotates) the disk. The normal speed of rotation is either 300 or 360 RPM(rotation per minute). This is low speed compared to hard disk. This is also why floppy has low data transfer rate. But the low speed makes it possible for the drive heads to be in physical contact with the disk without causing friction damage.

4) Circuit Board

A floppy disk drive incorporates logic boards. The logic board contains the circuitry used to control the head actuator, read-write heads, spindle motor, disk sensors and other components on the drive. The circuit on this board also talk to floppy disk controller over floppy interface. Theoretically, the logic board can be replaced or repaired.

5) The controller

Old days: the controller was an expansion card installed on ISA bus. It is a dedicated controller

Later on: a multifunction card that provides IDE/ATA, parallel port, serial port, and floppy interface.

Today: PCs have floppy controller integrated into the motherboard usually in the form of
super I/O chip. It includes parallel and serial ports too.

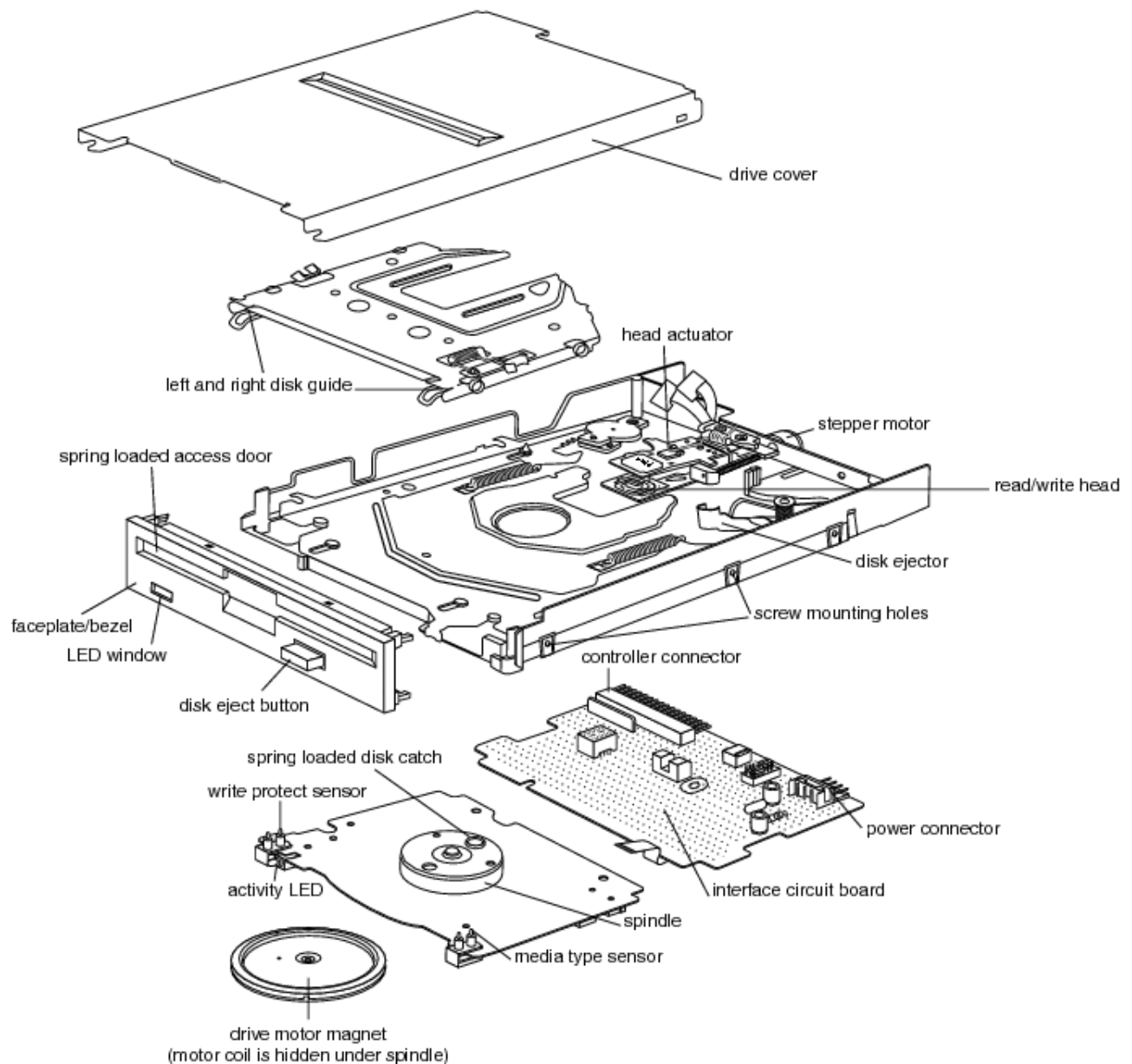
6) Connectors

Power Connector: 3¹/₂ floppy uses the mini connector for power

5¹/₄ floppy uses Molex connector.

Data Cable: Most floppy drives use the standard floppy interface. It uses 34 pin connector

where a ribbon cable is inserted. A ribbon cable connects floppy drive and
motherboard. It transfers data between them.



Rules for handling floppy disks:

Floppy disk can be damaged easily if we don't take care for them:

- ✓ Do not touch the recording surface with your fingers
- ✓ Do not bend the disk
- ✓ Do not spill coffee or other substances on the disk
- ✓ Do not put floppy near heat
- ✓ Do not put floppy near magnetic materials
- ✓ Do not write on the label with ball point pen
- ✓ Keep from dust, smoke, dirt, etc

Troubleshooting

1) The floppy LED remains on continuously.

Reason: data cable is backward either at the drive or controller side. Reinstall the cable.

2) Drive dead: the disk does not spin or LED is not on

Possible reasons:

- ✓ Bad power supply: measure the power at the cable using multimeter. 12v and 5v are expected to be supplied into the drive. If not, there is power problem.
- ✓ Bad data cable: the cable may be defective. Replace it and retest.
- ✓ Defective controller: install a card controller and retest

Hard Disk

Hard disk is the data center of PC. It is here that all your programs and data are stored.

A hard disk drive contains rigid, disk-shaped platters. The platters are usually constructed of aluminum or glass; so they can't bend or flex unlike floppy disk. The platters are coated by magnetic substance on both sides.

Data is written to hard disk by magnetizing and demagnetizing the surface of the platters. Data is stored on both sides of the platter.

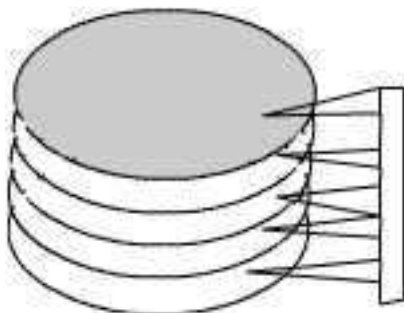


Fig 2 Hard disk platters with read-write head

The stack of platters form a hard disk. There is one head per platter side i.e. two head per platter. These heads do not touch the platters during normal operation. If forceful contact with spinning platters is created that is forceful enough to do damage, the event is called head crash. Dust and shock may cause head crash. Head crash could be catastrophic depending on situations → data loss, damage to heads, damage to surface of disk, etc.

During rest, the read-write head rests on the platters. It is during spinning that they should not touch.

Hard disk is also divided into tracks and sectors like floppy disk.

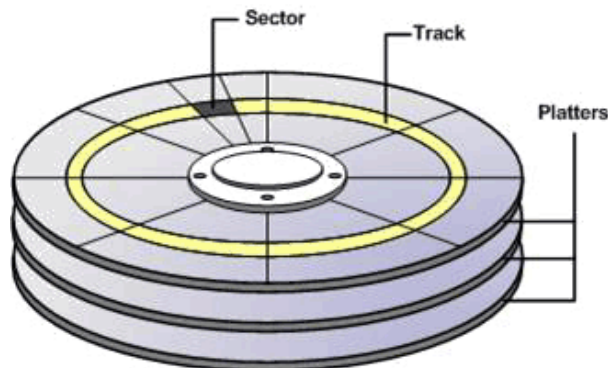


Fig 3 hard disk sectors and track

Most hard disk spin at 3,600 RPM—approximately 10 times faster than floppy disk. Some hard disks spin faster than this. Spinning rate affects data transfer rate.

Types of Hard Disks

Based on the connectors used to connect hard disks to motherboard, there are three types of hard disks:

- ✓ SCSI (Small Computer System Interface)
- ✓ SATA (Serial ATA)
- ✓ IDE (Integrated Drive Electronics) or ATA or PATA(Parallel ATA)

1) IDE

It is also called ATA(AT Attachment). In IDE, the interface electronics or controller is built into the drive itself; why it is called Integrated Drive Electronics. IDE is an evolutionary version of earlier interfaces that used separate drives and controllers.

Today, IDE is used to connect not only hard disk, but also CD-ROM drives, tape drives, and high capacity super Disk floppy drives.

AT Attachment with Packet Interface (ATA/ATAPI) is a standard interface used to connect storage devices such as hard disks, solid-state drives, and CD-ROM drives inside personal computers. ATA/ATAPI is the result of a long history of incremental technical development. With the market introduction of Serial ATA in 2003, the original ATA was retroactively renamed **Parallel ATA (PATA)**.

History and Terminology

The name of the standard was originally conceived as "AT Attachment" as its primary feature was a direct connection to the 16-bit ISA bus introduced with the IBM AT motherboard. The name was shortened to *AT Attachment(Advanced Technology Attachment)*.

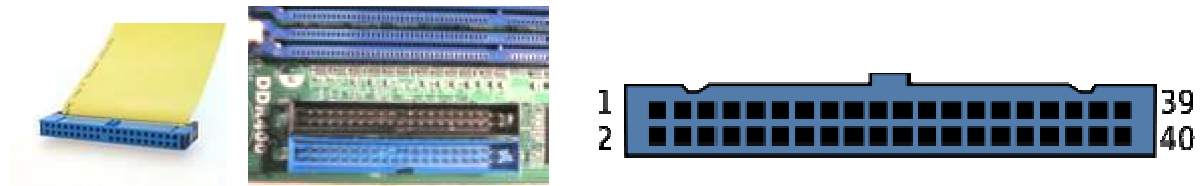


Fig ATA connector on the left and right, with two motherboard ATA connectors on the right.

It is designed by Western Digital and subsequently amended by many other companies.

Width: 16 bits

Bandwidth: 16 MB/s originally; later 33, 66, 100 and 133 MB/s

Max devices: 2 (one as master and the other as slave)

Connector: Parallel Cable 40 or 80 wires

IDE and ATA-1

The first version of what is now called the ATA/ATAPI interface was developed by Western Digital under the name Integrated Drive Electronics (IDE). The term Integrated Drive Electronics (IDE) refers not just to the connector and interface definition, but also to the fact that the drive controller is integrated into the drive, as opposed to a separate controller on or connected to the motherboard that is used prior to that. The integrated controller presented the drive to the host computer as an array of 512-byte blocks with a relatively simple command interface. The low-level details of the mechanical operation of the drive (e.g. moving the read write head) were now handled by the controller on the drive itself. This eliminated the need to design a single controller that could handle many different types of drives, since the controller could be unique for the drive. The host need only ask for a particular sector, or block, to be read or written, and either accept the data from the drive or send the data to it.

EIDE and ATA-2

In 1994, about the same time that the ATA-1 standard was adopted, Western Digital introduced drives under a slightly new name, **Enhanced IDE** (EIDE). These included most of the features of the forthcoming ATA-2 specification and several additional enhancements. Other manufacturers introduced their own variations of ATA-1 such as "Fast ATA" and "Fast ATA-2". ATA-2 was the first to note that devices other than hard drives could be attached to the interface.

ATAPI

As mentioned in the previous sections ATA was originally designed for and worked only with hard disks and devices that could emulate them. The introduction of ATAPI

(ATA Packet Interface) by a group called the Small Form Factor committee allowed ATA to be used for a variety of other devices that require functions beyond those necessary for hard disks. For example, any removable media device needs a "media eject" command, and a way for the host to determine whether the media is present, and these were not provided in the ATA protocol.

The Small Form Factor committee approached this problem by defining ATAPI, the "ATA Packet Interface". ATAPI is actually a protocol allowing the ATA interface to carry SCSI commands and responses; therefore all ATAPI devices are actually "speaking SCSI" other than at the electrical interface. In fact, some early ATAPI devices were simply SCSI devices with an ATA/ATAPI to SCSI protocol converter added on. The SCSI commands and responses are embedded in "packets" (hence "ATA Packet Interface") for transmission on the ATA cable. This allows any device class for which a SCSI command set has been defined to be interfaced via ATA/ATAPI.

ATAPI devices are being used to send the packets. On the other hand, ATA hard drives and solid state drives do not use ATAPI. ATAPI devices include CD-ROM and DVD-ROM drives, tape drives, and large-capacity floppy drives.

Drive size limitations

The original ATA specification used a 28-bit addressing mode, allowing for the addressing of 2^{28} (268,435,456) sectors (blocks) of 512 bytes each, resulting in a maximum capacity of 137 gigabytes (128 GB). ATA-6 introduced 48-bit addressing, increasing the limit to 128 PB (or 144 petabytes). As a consequence, any ATA drive of capacity larger than 137 gigabytes (128 GB; displayed by most operating systems as "128 GB") must be an ATA-6 or later drive. Connecting such a drive to a host with an ATA-5 or earlier interface will limit the usable capacity to the maximum of the controller.

Some OSs, including Windows 2000 pre-SP4, disable 48-bit LBA by default, requiring the user to take extra steps to use the entire capacity of an ATA drive larger than 137 gigabytes (128 GB).

Parallel ATA Interface

Until the introduction of Serial ATA, 40-pin connectors generally attached drives to a ribbon cable. Each cable has two or three connectors, one of which plugs into an adapter interfacing with the rest of the computer system. The remaining connector(s) plug into drives. Parallel ATA cables transfer data 16 bits at a time.

ATA's ribbon cables had 40 wires for most of its history (44 conductors), but an 80-wire version appeared with the introduction of the *Ultra DMA/33* (*UDMA*) mode. All of the additional wires in the new cable are ground wires, interleaved with the previously defined wires to reduce the effects of capacitive coupling between neighboring signal wires, reducing noise. Though the number of wires doubled, the number of connector pins and the pinout remain the same as 40-conductor cables,

and the external appearance of the connectors is identical. Internally the connectors are different.

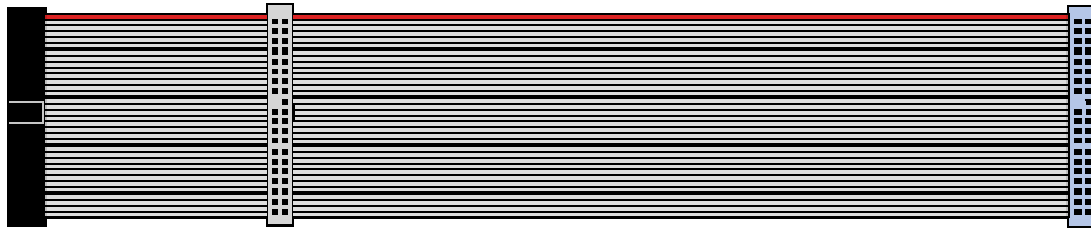
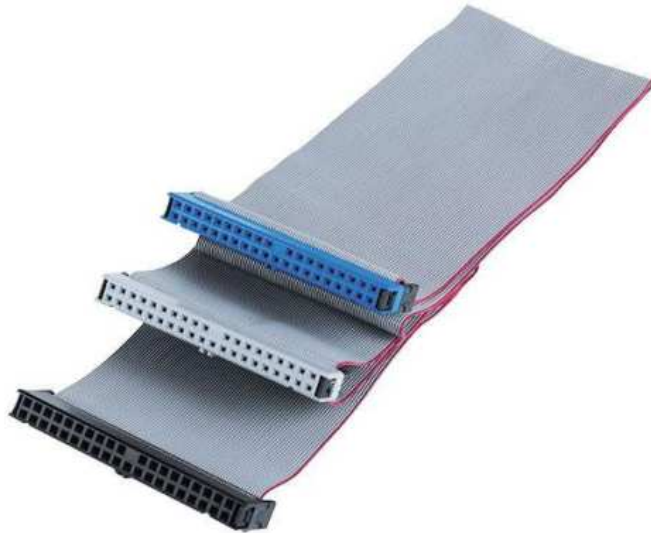


Fig. ATA cables: 40 wire ribbon cable, 80 wire ribbon cable (bottom)

Pin 20: In the ATA standard, Pin 20 is defined as key and is not used.

Multiple devices on a cable

It is possible to attach two devices to the same connector. In such cases, one is commonly referred to as a *master* and the other as a *slave*. The master drive generally appears first when the computer's BIOS and/or operating system enumerates available drives. On old BIOS (486 era and older), the drives are often referred to by the BIOS as "C" for the master and "D" for the slave following the way DOS would refer to the active primary partitions on each.

If there is a single device on a cable, it should be configured as *master*. However, some hard drives have a special setting called *single* for this configuration (Western Digital, in particular). Also, depending on the hardware and software available, a single drive on a cable can work reliably even though configured as the *slave* drive.

Cable select

A drive setting called *cable select* was described as optional in ATA-1 and has come into fairly widespread use with ATA-5 and later. A drive set to "cable select" automatically configures itself as master or slave, according to its position on the cable. Cable select is controlled by pin 28. The host adapter grounds this pin; if a device sees that the pin is grounded, it becomes the master device; if it sees that pin 28 is open, the device becomes the slave device.

This setting is usually chosen by placing a jumper on the "cable select" position, usually marked *CS*, rather than on the "master" or "slave" position. Note, pin 28 is only used by the drive to determine its position on the cable and thus whether it's the "master" or "slave"; it isn't in any way a "drive select" pin used when the host is issuing commands so if a drive is jumpered to "master" or "slave" its position on the cable doesn't have to match the pin 28 meaning of "master" and "slave".

HDD Passwords and Security

The disk lock is a built-in security feature in the disk. It is part of the ATA specification, and thus not specific to any brand or device. A disk always has two passwords: a User password and a Master password. Most disks support a Master Password Revision Code, which can tell you if the Master password has been changed, or if it still the factory default. The revision code is word 92 in the IDENTIFY response. A value of 0xFFFE means the Master password is unchanged.

ATA standards versions, transfer rates, and features

Congestion on the host bus to which the ATA adapter is attached may limit the maximum burst transfer rate. For example, the maximum data transfer rate for conventional PCI bus is 133 MB/s, and this is shared among all active devices on the bus.

Only the Ultra DMA modes use CRC to detect errors in data transfer between the controller and drive. This is a 16 bit CRC, and it is used for data blocks only.

Standard	Other Names	Transfer Modes (MB/s)	Maximum disk size	Other New Features
pre-ATA	IDE	PIO 0	2.1 GB	22-bit logical block addressing (LBA)
ATA-1	ATA, IDE	PIO 0, 1, 2 (3.3, 5.2, 8.3) Single-word DMA 0, 1, 2 (2.1, 4.2, 8.3) Multi-word DMA 0 (4.2)	137 GB	28-bit logical block addressing (LBA)
ATA-2	EIDE, Fast ATA,	PIO 3, 4: (11.1, 16.6)		PCMCIA connector

	Fast IDE, Ultra ATA	Multi-word DMA 1, 2 (13.3, 16.6)		
ATA-3	EIDE	Single-word DMA modes dropped		S.M.A.R.T., Security
ATA/ATAPI-4	ATA-4, Ultra ATA/33	Ultra DMA aka UDMA/33		AT Attachment Packet Interface (ATAPI) (support for CD-ROM, tape drives etc.), Optional overlapped and queued command set features
ATA/ATAPI-5	ATA-5, Ultra ATA/66	Ultra DMA aka UDMA/66		80-wire cables; CompactFlash connector
ATA/ATAPI-6	ATA-6, Ultra ATA/100	UDMA 5 aka UDMA/100	144 PB	48-bit LBA
ATA/ATAPI-7	ATA-7, Ultra ATA/133	UDMA 6 aka UDMA/133 SATA/150		SATA 1.0, Streaming feature set, long logical/physical sector feature set for non-packet devices
ATA/ATAPI-8	ATA-8	—		Hybrid drive featuring non-volatile cache to speed up critical OS files

2) SATA

Serial Advanced Technology Attachment (SATA) is a computer bus primarily designed for transfer of data between a computer and mass storage devices such as hard disk drives and optical drives.

The main advantages over the older parallel ATA interface are faster data transfer, ability to remove or add devices while operating (hot swapping), thinner cables that let air cooling work more efficiently, and more reliable operation with tighter data integrity checks.

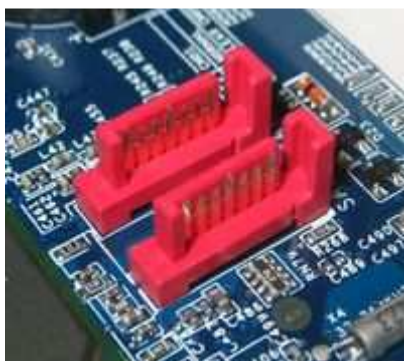


Fig SATA connector on motherboard

It was designed as a successor to the Advanced Technology Attachment standard (ATA), and is expected to eventually replace the older ATA technology. Serial ATA adapters and devices communicate over a high-speed serial cable.

Advanced Host Controller Interface

The standard interface for SATA controllers is Advanced Host Controller Interface (AHCI), which allows advanced features of SATA such as hot plug and Native Command Queuing (NCQ). If AHCI is not enabled by the motherboard and chipset, SATA controllers typically operate in IDE emulation mode which does not allow features of devices to be accessed that are not supported by the ATA/IDE standard. Windows device drivers that are labeled as SATA are usually running in IDE emulation mode unless they explicitly state that they are AHCI.

Throughput

a) SATA 1.5 Gbit/s

First-generation SATA interfaces, also known as SATA/150 or unofficially as SATA 1, communicate at a rate of 1.5 gigabits per second (Gbit/s). Data transfer rates are limited by mechanical hard drives themselves, not the interfaces: the fastest modern desktop hard drives transfer data at a maximum of about 118 MB/s, which is well within the capabilities of even the older PATA/133 specification.

b) SATA 3.0 Gbit/s

Soon after SATA/150s introduction a number of shortcomings were observed. At the application level SATA could only handle one pending transaction at a time like PATA. The SCSI interface has long been able to accept multiple outstanding requests and service them in the order which minimizes response time. This feature, Native Command Queuing (NCQ), was adopted as an optional supported feature for SATA 1.5 Gbit/s and SATA 3.0 Gbit/s devices.

Backward compatibility between SATA 1.5 Gbit/s controllers and SATA 3.0 Gbit/s devices was important, so SATA/300's auto negotiation sequence is designed to fall back to SATA/150 speed (1.5 Gbit/s rate) when in communication with such devices.

Cables and Connectors

Connectors and cables are the most visible difference between SATA and Parallel ATA drives. SATA power connectors and data connectors have been criticized for their fragility and poor robustness — the thin plastic tops of the connectors can easily break due to shearing force when the user pulls the plug at a non-orthogonal angle, as can the connectors on drives they connect to. In the case of a broken connector on a hard drive, this could result in a complete loss of access to all data stored on the drive.

a) Data Cable

The SATA standard defines a data cable with seven conductors and wafer connectors on each end. SATA connectors and cables are easier to fit in closed spaces and reduce obstructions to air cooling. They are more susceptible to accidental unplugging and breakage than PATA.



Parallel ATA uses single-ended signaling which results in the noise combining with the data signal during the signal transmission. Noise causes significant interference with the data signal at higher speeds. In comparison, SATA systems use differential signaling whereby it is easy to filter out the noise from the data signal at the receiving end. The higher noise rejection allows the SATA system to carry the signal at higher speeds without distortion or noise interference.

b) Power Connector

The SATA standard also specifies a new power connector. Like the data cable, it is wafer-based, but its wider 15-pin shape prevents accidental misidentification and forced insertion of the wrong connector type. Native SATA devices favor the SATA power-connector over the old four-pin Molex connector, although some SATA drives retain older 4-pin Molex in addition to the SATA power connector.

There are more pins than the traditional connector for several reasons:

- A third voltage is supplied – 3.3 V – in addition to the traditional 5V, and 12V.
- Each voltage is supplied by three pins ganged together – because the small pins by themselves cannot supply sufficient current for some devices. (Each pin should be able to provide 1.5 A.)
- For each of the three voltages, one of the three pins is used for hotplugging.
- Ground is provided by five pins ganged together.
- Pin 11 can be used for staggered spinup, activity indication, or nothing. Staggered spinup is used to prevent many drives from spinning up simultaneously, as this may draw too much power. Activity is an indication of whether the drive is busy, and is intended to give feedback to the user through an LED.

Pin #	Function
1–3	3.3Volts
4–6	Ground
7–9	5V
10	Ground
11	Staggered up/Activity spin
12	Ground
13–15	12V



A 15-pin Serial ATA power connector.

Adaptors are available to convert a 4-pin Molex connector to a SATA power connector. However, because the 4-pin Molex connectors do not provide 3.3 V power,

these adapters provide only 5 V and 12 V power and leave the 3.3 V lines unconnected.

SATA and SCSI

SCSI currently offers transfer rates higher than SATA, but is a more complex bus usually resulting in higher manufacturing cost. Some drive manufacturers offer longer warranties for SCSI devices, however, indicating a possibly higher manufacturing quality control of SCSI devices compared to PATA/SATA devices. SCSI buses also allow connection of several drives (using multiple channels, 7 or 15 on each channel), whereas SATA allows one drive per channel, unless using a port multiplier.

SATA 3.0 Gbit/s offers a maximum bandwidth of 300 MB/s per device compared to SCSI with a maximum of 320 MB/s. Also, SCSI drives provide greater sustained throughput than SATA drives because of disconnect-reconnect and aggregating performance.

SCSI drives are typically more expensive so they are traditionally used in servers and disk arrays where the added cost is justifiable. Inexpensive ATA and SATA drives evolved in the home computer market, hence the general opinion is that they are less reliable. As those two worlds started to overlap, the subject of reliability became somewhat controversial.

Unlike PATA, both SATA and eSATA are designed to support hot-swapping-the process of connecting or removing SATA devices while the machine is running. However, this feature requires proper support at the host, device (drive), and operating-system level.

SCSI devices with SCA-2 connectors are designed for hot-swapping. Many server and RAID systems provide hardware support for transparent hot-swapping. The SCSI standard prior to SCA-2 connectors was not designed for hot-swapping, but, in practice, most RAID implementations support hot-swapping of hard disks.

3) SCSI

The Small Computer System Interface (SCSI, pronounced as scuzzy) standard, like the IDE concept, provides a true system level interface for the drive. Nearly all the drive's controller electronics are located on the peripheral device. As with the IDE host adapter, the duties of the SCSI host adapter are reduced to mostly physical connection functions, along with some signal compatibility handling.

It is a technology which allows to connect different hardware devices such as hard disk drives, CD-ROM drives, tape streamers and scanners to a single adapter. Most systems support upto 4 host adapters, each with upto 15 devices.

SCSI is a high end technology. It is a fast interface generally suited to high performance machines. It is especially used in high end PC's such as network servers,

workstations, or anywhere where storage performance is needed. SCSI might be compared to the EIDE interface, which also uses a host adapter controlling drives.

However SCSI have two major advantages to EIDE:

- ✓ A SCSI host controls 7 or 15 devices (using only one IRQ).
- ✓ The SCSI system holds its own computer power, thus freeing the CPU from workload. If you are critical about your PC power, the SCSI would be worth considering.

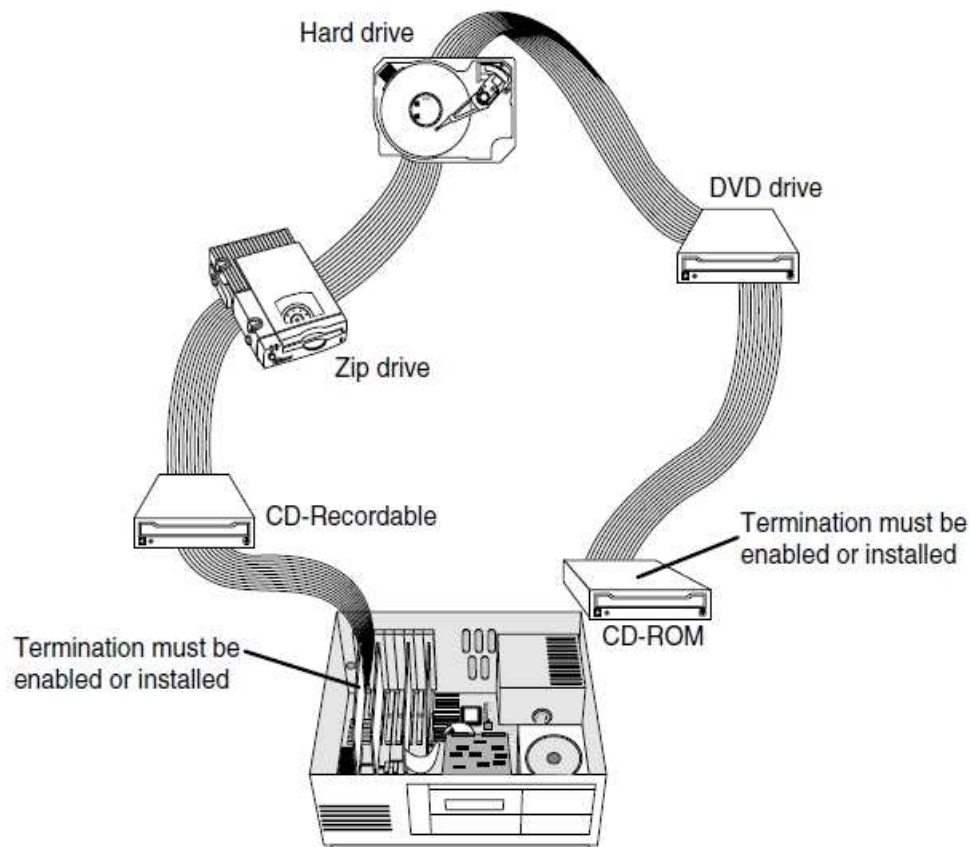


Fig SCSI drive

The host adapter

A SCSI system is built around a central, intelligent controller called the host adapter. A host adapter can control several SCSI units:

- ✓ Many units on the same host adapter.
- ✓ Many types of drives: Hard disks, CD ROM's, Zip drives, CDROM recorders etc.
- ✓ Tape-streamers (DAT and others).
- ✓ Scanners.

The host adapter has its own BIOS separate from the PC BIOS. When you boot the PC, you will see the adapter communicating with connected SCSI devices.

The SCSI interface can be used to connect up to seven (or 15) diverse types of peripherals to the system. As an example, a SCSI chain could connect a controller to a hard drive, a CD-ROM drive, a high-speed tape drive, a scanner, and a printer. Additional SCSI devices are added to the system by daisy-chaining them together.

SCSI is Intelligent

SCSI is remarkable in having an intelligent protocol, which assures maximum utilization of the data transfer. The basis of SCSI is a set of *commands*. Each individual device holds its own controller, which interprets these commands. All commands within the SCSI system are handled internally, meaning the CPU does not have to control the process.

In PC-compatible systems, the SCSI interface uses a 50-pin signal cable arrangement. Although the SCSI interface defined here is the full standard, some manufactures may not utilize all the functions of the interface.

When installing a SCSI device, each device needs SCSI ID. SCSI ID was originally set by jumpers on the host adapter card. In PnP systems, the BIOS configure the address using information obtained directly from the card during boot up. Unlike other HDD types, SCSI hard drives are not configured as part of the system's CMOS setup function.

Each SCSI device has either a SCSI number selection switch or configuration jumpers. The SCSI address setting must be unique for every device attached to the host adapter. If two devices are set to the same ID number, one or both of them will appear invisible to the system.

The SCSI daisy chain must be terminated with a terminating resistor network pack at both of its ends. Single-connector SCSI devices are normally terminated internally. If not, a SCSI terminator cable (containing a built-in resistor pack) must be installed at the end of the chain. SCSI termination is a major cause of SCSI-related problems. Poor terminations cause a variety of different system problems including failed system startups, hard drive crashes, and random system failures.

The increased speed capabilities of the SCSI interfaces make them attractive for intensive applications such as large file servers for networks, and multimedia video stations. The EIDE interface is generally more widely used, however, because of its lower cost and nearly equal performance. The EIDE device also tends to be more compatible with DOS. Most SCSI adapters come with software drivers that must be installed to support them.

Optical Disks

A typical CD-ROM disk is 4.7 inches in diameter, and consists of three major parts:

- ✓ Acrylic substrate
- ✓ Aluminized, mirror-finished data surface
- ✓ Lacquer coating

The scanning laser beam comes up through the disk, strikes the aluminized data surface, and is reflected back. Because there is no physical contact between the reading mechanism and the disk, the disk never wears out. This is one of the main

advantages of the CD system. The blisters on the data surface are typically just under 1 micrometer in length, and the tracks are 1.6 micrometers apart. The data is encoded by the length and spacing of the blisters and the lands between them.

With a CD, data is written digitally on a light-sensitive material by a powerful, highly focused laser beam. The writing laser is pulsed by the modulated data to be stored on the disk. When the laser is pulsed, a microscopic blister is burned into the optical material, causing it to reflect light differently from the material around it. The blistered areas are referred to as pits; the areas between them are called lands.

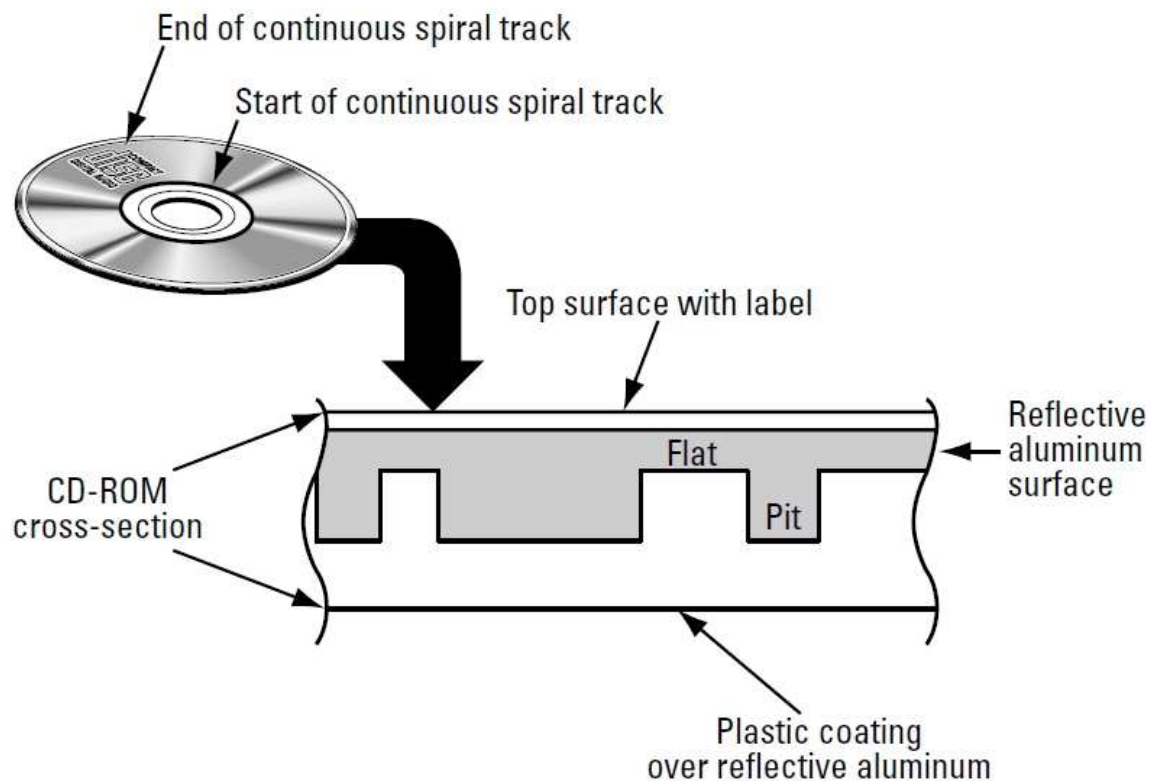


Fig CD-ROM data writing mechanism

The recorded data is read from the disk by scanning it with a lower-power, continuous laser beam. The laser diode emits the highly focused, narrow beam that is reflected back from the disk. The reflected beam passes through a prism and is bent 90 degrees, where it is picked up by the diode detector and converted into an electrical signal. Only the light reflected from a land on the disk is picked up by the detector. Light that strikes a pit is scattered and is not detected. The lower power level used for reading the disk ensures that the optical material is not affected during the read operation.

With an audio CD, the digital data retrieved from the disk is passed through a digital-to-analog converter (DAC) to reproduce the audio sound wave. This is not required for digital computer systems, however, because the information is already in a form acceptable to the computer. Therefore, CD players designed for use in computer systems are referred to as CD-ROM drives, to differentiate them from audio CD players. Otherwise, the mechanics of operation are very similar between the two

devices. The ROM designation refers to the fact that most of the drives available are read-only.

The overall operation of CD-ROM drive is as follows:

1. The laser diode emits a low-energy infrared beam toward a reflecting mirror
2. The servo motor positions the beam onto the correct track on the CD-ROM by moving the reflecting mirror
3. When the beam hits the disc, its refracted light is gathered and focused through the first lens beneath the platter, bounced off the mirror, and sent toward the beam splitter
4. The beam splitter directs the returning laser light toward another focusing lens
5. The last lens directs the light beam to a photo detector that converts the light into electric impulses
6. These incoming impulses are decoded by the microprocessor and sent along to host computer as data

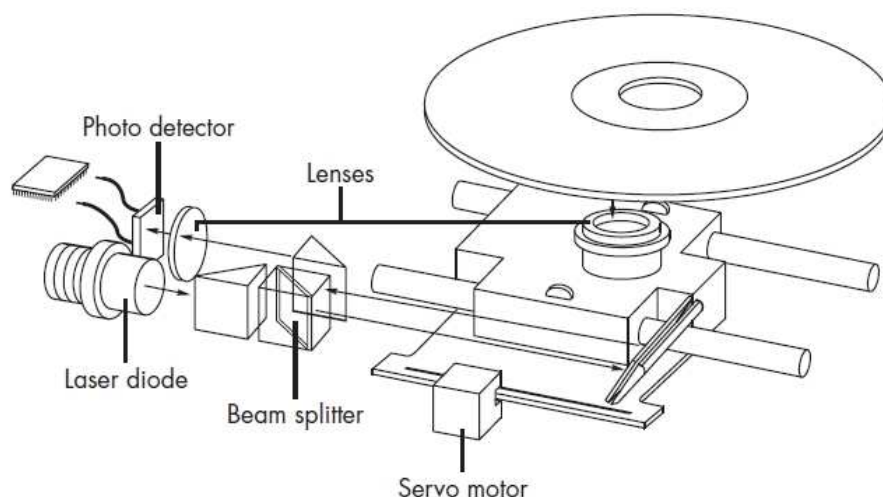


Fig CD-ROM drive components

Blu-Ray Disc

Blu-ray Disc is a high-capacity CD/DVD-type optical disc format.

It can store data upto 27GB or 2 hours of high-definition video.

Blu-ray discs will be contained in a cartridge that protects them from dust and fingerprints.

It has the same size as CD/DVD.